MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION OCTOBER 2025

THIRD SEMESTER

Discipline Specific Core Course (DSC)-MG3DSCECO200 –

MICROECONOMICS I

(2024 ADMISSION ONWARDS)

SCHEME OF VALUATION

Duration: 2 Hours **Maximum Marks:70**

Part A

1. b, Strong ordering

2. a, Isoquant is tangent to isocost

3. d, Indifference curve analysis

4. c, Equal to AC

5. c, The consumer's real income changes

6. c, Marginal products of inputs

7. a, MR=MC

8. b, Dumping

9. c, Richard Stone

10. b, Economic region of production

Part B

11. First condition -Marginal Rate of Substitution (MRS) between two goods must equal the ratio of their prices.

Second condition-The second condition for equilibrium is that the chosen point must be on the highest attainable indifference curve, given the budget constraint. The indifference curve must also be convex to the origin at the point of tangency, which means the MRS must be diminishing.

12. Isoquants slope downward, Isoquants are convex to the origin, Higher isoquants indicate greater output, Isoquants never intersect, Isoquants do not touch the axes, Isoquants are not always parallel.

13. The firm will produce and sell at a point where the price covers at least SAVC of production. If the price falls below SAVC, the firm will shut down and exit the industry. The shut down point is that output level where P=SAVC.

14. The primary objective of a cartel is to maximize the joint profits of its member firms by reducing competition and manipulating market conditions, typically through controlling output, fixing prices, or dividing markets among themselves.

15. In the long run, a perfectly competitive market eliminates both supernormal profits and losses due to the mechanisms of free entry and exit of firms in the industry. When existing firms earn supernormal profits, new firms are attracted to enter the market. This increases the total supply of the good, leading to a decrease in market price. Entry continues until all firms are only making normal profits, where price equals average cost, and supernormal profits are no longer possible.

If firms are incurring losses in the short run, some will eventually exit the industry. As firms leave, the supply decreases, causing the market price to rise. Exit continues until the remaining firms are also only making normal profits—where price once again equals average cost, and further losses are eliminated

16. Imperfect competition refers to market structures where the assumptions of perfect competition—such as identical products, unlimited sellers and buyers, perfect information, and costless market entry/exit—are not fully met.eg: Monopoly, Oligopoly, Duopoly, Monopolistic competition, Monopsony

17. The optimal input combination refers to the most efficient mix of inputs—such as labour and capital—that enables a firm to produce a given level of output at the lowest possible cost or, alternatively, maximize output for a given expenditure on inputs.

Conditions -Isoquant tangency with Iso-cost line, Equality of MRTS and Input price ratio

18. Monopoly is considered less efficient than perfect competition because monopolies restrict output and charge higher prices, resulting in both allocative and productive inefficiency. In perfect competition, firms are price takers and produce where price equals marginal cost, making sure that resources are allocated to their most valued use and maximizing total surplus for society. In monopoly, seller sets a higher price by restricting output to maximize profit, causing the price to exceed marginal cost.

19. Excess capacity becomes a waste when resources such as capital, labour, and raw materials are underutilized in production, leading to higher average costs and inefficiencies for firms and the economy. In economic terms, it means firms operate below their optimal production levels, failing to minimize average costs and thereby squandering potential output that could benefit society.

20. The convexity of an indifference curve is primarily due to the law of diminishing marginal rate of substitution (MRS). As a consumer substitutes one good for another, the amount of the second good they are willing to give up to gain additional units of the first good decreases, assuming both goods are imperfect substitutes.

21. The Giffen paradox in Economics describes a rare situation where the demand for a good increases as its price rises, directly contradicting the standard law of demand that predicts a negative relationship between price and quantity demanded. This phenomenon typically occurs with certain basic staple goods known as Giffen goods, which are usually consumed by lower-income groups and have few or no close substitutes.

22. The economic region of production refers to the range of input combinations—typically labour and capital—within which a firm can operate efficiently, ensuring both inputs have a positive marginal product and contribute meaningfully to output. The region is bounded by ridge lines on an isoquant map, which mark the points where the marginal productivity of an input becomes zero, indicating the boundary between rational (efficient) and irrational (inefficient) production zones.

23. Price discrimination is a pricing strategy where a seller charges different prices for the same product or service to different consumers or groups, not due to cost differences but based on varying willingness or ability to pay. This allows firms to

capture more revenue and reduce consumer surplus.

24. Pure (Perfect) Oligopoly: All firms produce homogeneous products, such as steel or cement. Imperfect (Differentiated) Oligopoly: Firms produce differentiated products, such as automobiles or soft drinks. Collusive Oligopoly: Firms cooperate—either explicitly or tacitly—to set prices, output, or both, often forming cartels like OPEC. Non-Collusive Oligopoly: Firms compete and do not cooperate, resulting in strategic behaviour and price rigidity, as seen in the airline industry.

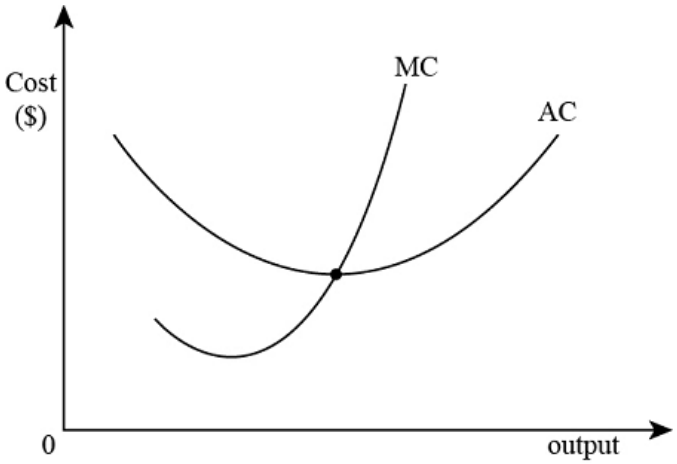
25. The pragmatic approach involves developing correlations and solutions based on empirical data and observable market phenomena. n areas such as demand analysis, pragmatists will construct demand functions directly from market data rather than from models grounded in consumer utility or introspection.

Linear expenditure system are aggregate econometric models that deal with consumer expenditure of groups of commodities. Here demand functions are derived from utility maximization functions subject to the budget constraint.

Part C

26. Limitations- Considers only two goods - assumes consumers behave rationally and consistently - sometimes overlook the importance of budget constraints and changes in income or wealth - oversimplification of income and substitution effects - practical limitations of ordinal measurement - ignores psychological and sociological factors - assumes convex preferences, but actual preferences may be non-convex.

27.



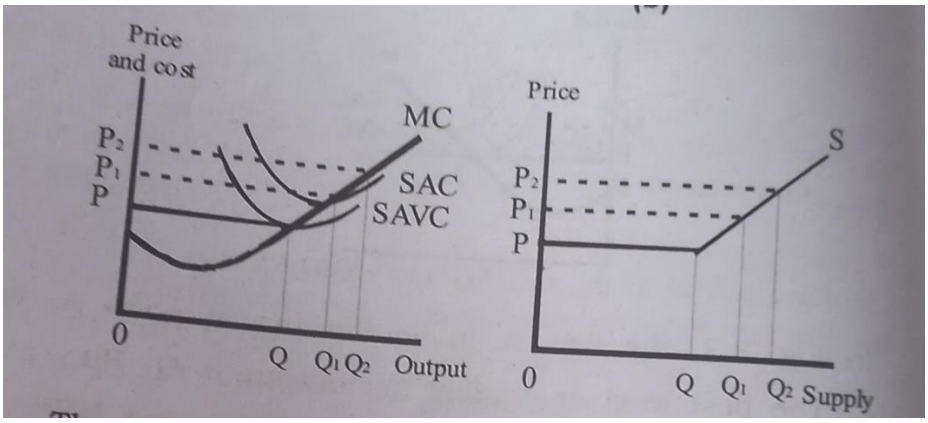
When MC is below AC, the AC is falling as output increases.

When MC is above AC, the AC is rising as output increases

MC equals AC at the minimum point of the AC curve indicates the most efficient scale of production

(Diagram- 2 marks, Explanation-4 marks)

28. The industry supply curve in perfect competition is derived by horizontally summing the individual supply curves of all the firms in the industry. Each firm's supply curve is the part of its marginal cost (MC) curve that lies above its average variable cost (AVC) curve. This shape arises because firms supply output where price equals marginal cost (P = MC) as long as price covers average variable cost.

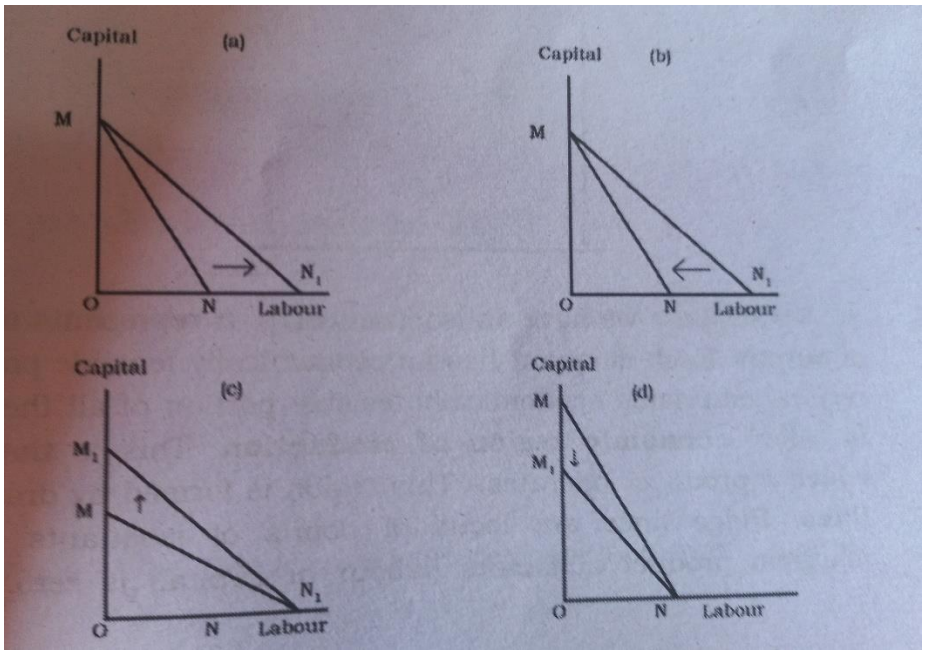


(Diagram- 2 marks, Explanation-4 marks)

29. New welfare economics differs from traditional welfare economics primarily in its approach to measuring and comparing welfare. New welfare economics avoids interpersonal comparisons of utility and instead focuses on efficiency criteria like Pareto efficiency. It uses a formal, mathematical framework emphasizing when an allocation is efficient and leaves questions of equity or fairness to political or social considerations. In contrast, traditional welfare economics assumed cardinal utility—that individual utilities could be measured and compared—and focused on constructing a social welfare function by summing individual utilities, considering distributive efficiency closely connected to utility comparisons.

30. An iso-cost line represents all the combinations of two inputs, typically labour and capital, that a firm can purchase with a given total budget or cost outlay. The concept is similar to a consumer's budget line but applies to production inputs rather than goods. This change in input price does not cause a parallel shift but a rotation

(swing) of the iso-cost line about one axis, altering the slope to reflect the new ratio of input prices.



a. Wage rate falls,
b. Wage rate rises
c. Interest rate falls
d. Interest rate rises

(Iso cost line explanation – 4 marks, For any one diagram mentioned above, give 2 marks)

31. A budget line is a graphical representation showing all possible combinations of two goods that a consumer can buy with a given income and the prices of those goods, such that the total expenditure equals the consumer's income. It reflects the consumer's purchasing power and trade-offs between the two goods given their budget constraint.

Importance

It helps consumers make informed choices by showing affordable combinations of goods, aiding in maximizing satisfaction within a limited income.

Budget line illustrates the spending limit, preventing overspending

It plays a key role in consumer theory, working with indifference curves to find the consumer's optimal choice.

Changes in income or prices shift the budget line, helping understand how buying

patterns adjust.

Slope of the budget line indicates opportunity cost — what must be given up to obtain more of the other good.

(Budget line explanation- 3 marks, Importance of budget line- 3 marks)

32. **Value liberally** -The concept of reserve capacity in modern cost theory is important because it introduces flexibility into the production process, allowing firms to produce a range of output levels at low or constant cost without the immediate need for additional investment. Reserve capacity is the planned built-in capability of a plant or firm to handle fluctuations in demand or temporary increases in output.

Changes in shape of short run cost curves

Changes average variable cost curve from U-shaped to saucer-shaped with a flat region.

Short-run marginal cost curve is flat and coincides with the average variable cost in the reserve capacity range.

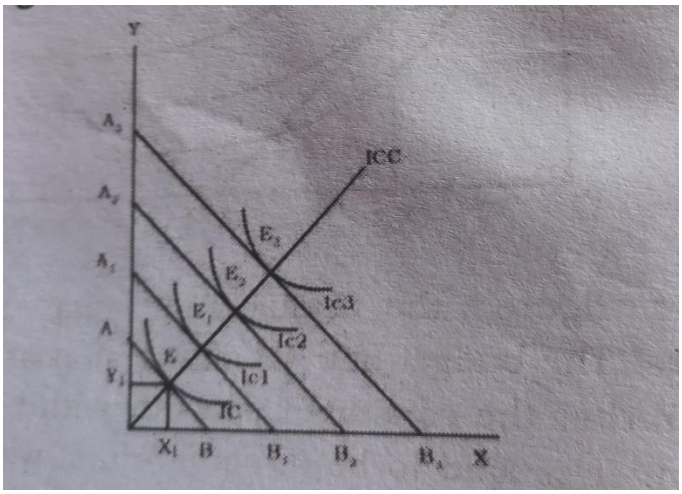
Short-run average cost curve falls, flattens due to reserve capacity, then rises after reserve capacity is used up.

33. In a monopoly, one firm supplies a particular good or service and can set prices without competition from other sellers. This seller has market power, usually resulting in higher prices and fewer choices for consumers. Monopolies are regulated by antitrust laws to prevent abuse of power and promote competition.

In a monopsony, one buyer dominates the market for a good or service, controlling its demand. The buyer can influence prices, often resulting in lower prices or wages for sellers or workers. Monopsonies often lead to market inefficiencies, such as lower wages in labor markets.

34. **Value liberally** -The bargaining power determines where the price settles between the two extremes—the monopolist’s price (higher) and the monopsonist’s price (lower). If the monopolist has greater bargaining power, the price tends to be closer to the monopolist's preferred higher price. Conversely, if the monopsonist holds more power, the price moves closer to the lower monopsony price. When the bargaining power is roughly equal, the price settles in between, potentially leading to a "win-win" equilibrium where both sides share the profit fairly.

35.



If a person's income increases, the income consumption curve (ICC) shows how their consumption choices change when prices remain constant. The budget line shifts outward parallel to the original budget line, reflecting the higher income. With increased income, the consumer moves to higher indifference curves, enjoying greater satisfaction and consuming larger quantities of goods if they are normal goods.

(Diagram- 2 marks, Explanation- 4 marks)